MILITARY SPECIFICATION SHEET

SWITCHES, MULTISTATION, PUSH BUTTON, ILLUMINATED MATRIX ARRANGEMENT PLUG-IN MODULE, 2 AMPERES, UNSEALED, TYPE SM09L1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24317.

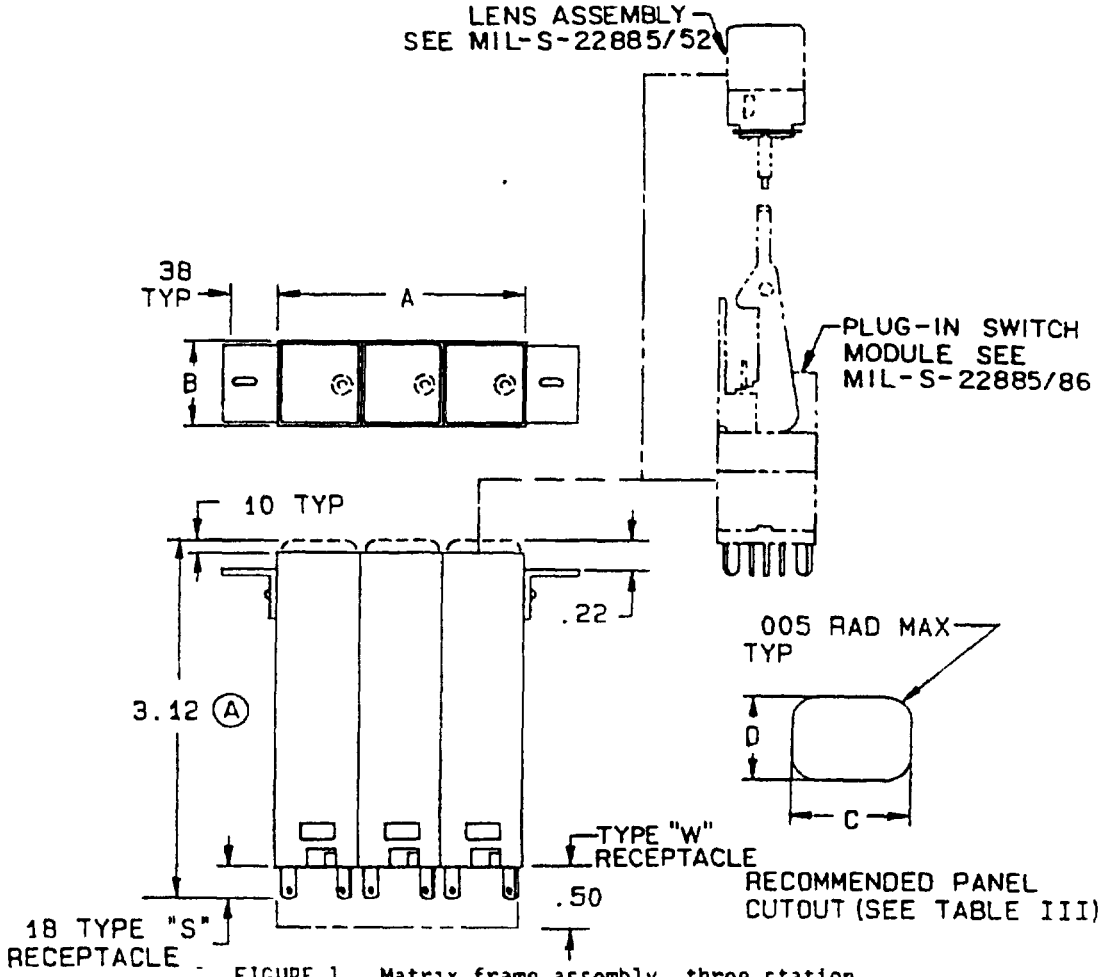
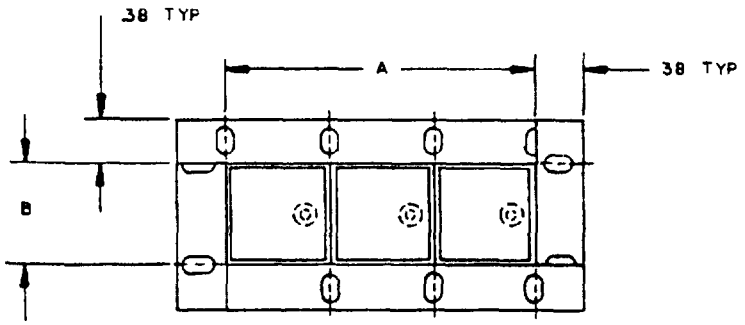


FIGURE 1. Matrix frame assembly, three station.

Ⓐ denotes changes

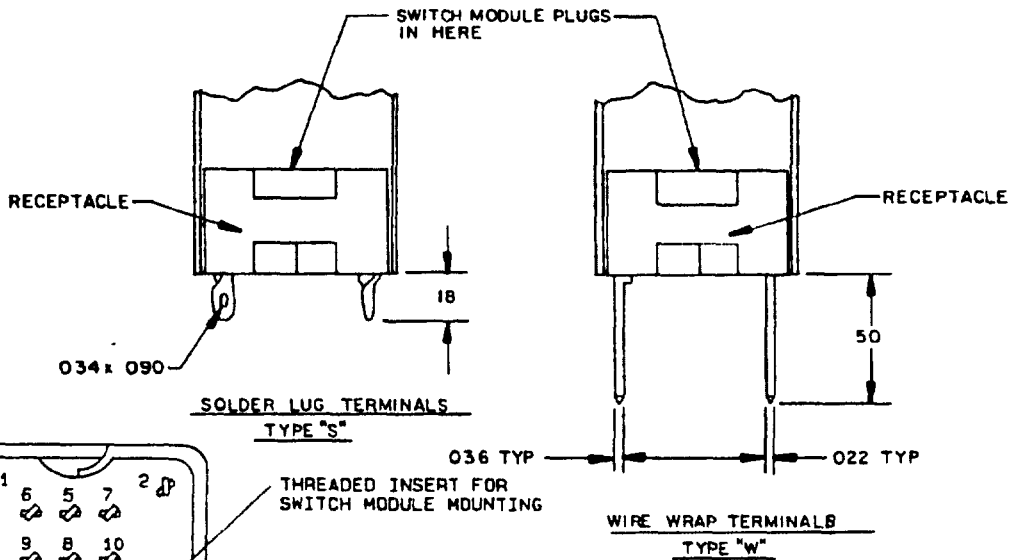
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

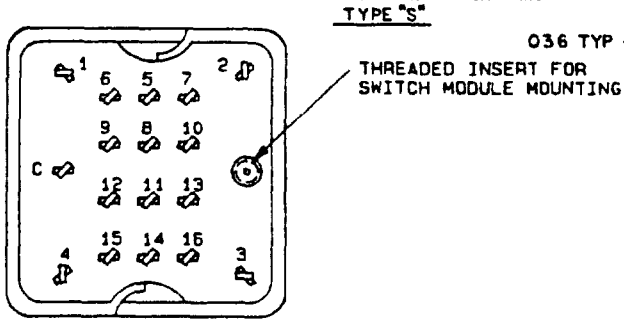
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Inches	mm
.005	0.13
.022	0.56
.034	0.86
.036	0.91
.090	2.29
.10	2.5
.18	4.6
.22	5.6
.38	9.7
.50	12.7
3.12	79.2

FLANGE ARRANGEMENT FOR RFI SHIELDING



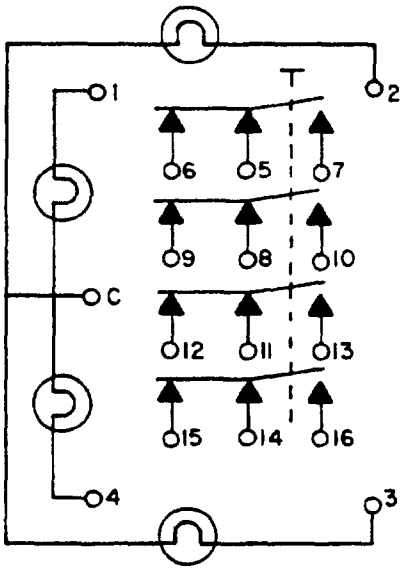


TERMINAL ARRANGEMENT

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.
4. Terminals shall be permanently identified with numbers as shown in terminal arrangement.
5. Design configuration is optional within envelope dimensions.
6. For dimensions, see table III.

FIGURE 1. Matrix frame assembly, three station - Continued.

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NOTES:
1. Terminals 8 through 13 are omitted for 2PDT configurations.
2. Terminals 5 through 16 are omitted for indicator configurations.
3. Terminals 5, 8, 11, and 14 are pole terminals.

FIGURE 2. Lamp and 4PDT switch schematic.

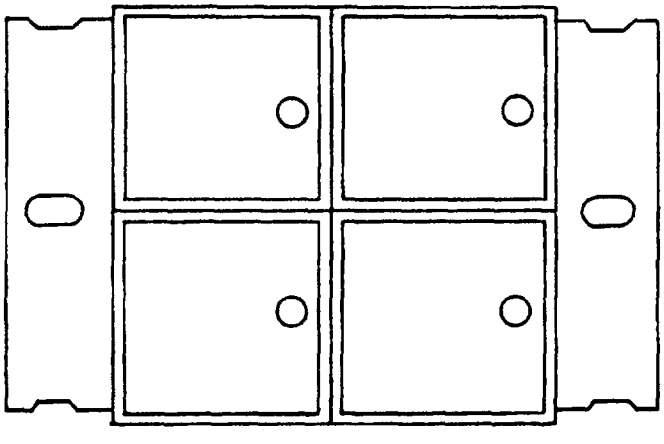


FIGURE 3. Flange arrangement for unshielded switch (vertical flanges shown).

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REQUIREMENTS:

Ⓐ Dimensions and configuration. See figures 1 through 3.

Complete multistation switch assembly shall consist of:

a. Multistation matrix frame (includes receptacles and flanges).

b. Switch module, plug-in type (see MIL-S-22885/86), when specified in contract or purchase order.

c. Lamps (T 1-3/4 midget flange base incandescent), when specified in contract or purchase order.

d. Pushbutton lens (see MIL-S-22885/52), when specified in contract or purchase order.

Ⓐ Indication characteristic: L (Illuminated).

Ⓐ Enclosure design: 1 (Unsealed).

Frame finish.

Non-RFI shielded. Black anodize in accordance with MIL-A-8625.
RFI shielded: Chemical filmed in accordance with MIL-C-5541.

RFI attenuation Requirement and test in accordance with MIL-S-22885/86.

Weight. 1.32 ounces (approximate) per station (37.4 grams).

Electrical ratings See table I.

TABLE I. Electrical ratings.

Load	Sea level	
	28 V dc	115 V ac
	(amperes)	(amperes)
Resistive	2.0	2.0
Inductive	1.0	1.0
Lamp	0.75	0.75

Operating force. 3.5 ±1.0 pounds.

Colors and photometric brightness. In accordance with MIL-S-22885/52.

Ⓐ Solderability: Not applicable to wire wrap terminals.

Shock: Test condition A, method 213 of MIL-STD-202 (50 g, half-sine).

Matrix frame part number: M24317/9- (dash number from table III).

Qualification inspection: All applicants for qualification approval shall demonstrate that each of their items conforms to all the requirements specified in the applicable documents, singularly and in combination with all other previously qualified items, regardless of manufacturer. When the lens assembly and plug-in switch modules are qualified to MIL-S-22885/52 and /86, qualification and group C inspection of the multistation switch assembly shall be in accordance with table II.

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Extent of qualification: The multistation switch and indicator assembly to be used for qualification inspection shall be the 3-station unit (3 x 1 matrix). Qualification of the 3-station units shall qualify all multistation units up to and including the 20-station unit. The 3-station units shall contain the following items. Locations of each item in the matrix will be determined by the test activity.

Item	Part number	Description	Quantity
Ⓐ 1	M24317/9-023	3 x 1 matrix frame (RFI shielded)	1
2	M22885/86-003	2PDT momentary switch module	1
3	M22885/86-001	4PDT momentary switch module	1
4	M22885/86-005	Indicator module	1
5	M22885/52-	Pushbutton lens	3

TABLE II. Qualification inspection (group submission).

Assembly	Qualification inspection table of MIL-S-24317		Extent of approval
	Group	Number of sample units	
1 x 3 Matrix (RFI shielded)	I 1/	8	All
	II 2/	4 from group I	
	III 3/	2 from group II 4/	

1/ Simultaneity not applicable.
2/ Torque strength of plunger not applicable.
3/ Dielectric withstanding voltage after salt spray not applicable.
4/ Samples shall be subjected to RFI test of MIL-S-22885/86.

Ⓐ TABLE III. Dash number, design and panel cutout dimensions.

Dash number 1/		Number of stations (H x V)	Dimensions 2/				Number of flanges 3/	RFI shielded
			Matrix ±.015 (.38)		Panel cutout +.030 (.76) -.000			
Type S	Type W		A	B	C	D		
-001	-011	1 (1 x 1)	.755 (19.18)	.755 (19.18)	.775 (19.69)	.775 (19.69)	2V	No
-002	-012	2 (2 x 1)	1.510 (38.35)	"	1.530 (38.86)	"	2V	No
-003	-013	3 (3 x 1)	2.265 (57.53)	"	2.285 (58.04)	"	2V	No
-004	-014	4 (4 x 1)	3.020 (76.71)	"	3.040 (77.22)	"	2V	No
-005	-015	5 (5 x 1)	3.775 (95.89)	"	3.795 (96.39)	"	2V	No
-006	-016	6 (6 x 1)	4.530 (115.06)	"	4.550 (115.57)	"	2V	No
-007	-017	7 (7 x 1)	5.285 (134.24)	"	5.305 (134.75)	"	2H	No
-008	-018	8 (8 x 1)	6.040 (153.42)	"	6.060 (153.92)	"	2H	No

See footnotes at end of table.

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Ⓐ TABLE III. Dash number, design and panel cutout dimensions - Continued.

Dash number 1/		Number of stations (H x V)	Dimensions 2/				Number of flanges 3/	RFI shielded
			Matrix ±.015 (.38)		Panel cutout +.030 (.76) -.000			
Type S	Type W		A	B	C	D		
-009	-019	9 (9 x 1)	6.795 (172.59)	.755 (19.18)	6.815 (173.10)	.775 (19.69)	2H	No
-010	-020	10 (10 x 1)	7.550 (191.77)	"	7.570 (192.28)	"	2H	No
-021	-031	1 (1 x 1)	.755 (19.18)	"	.775 (19.69)	"	2V2H	Yes
-022	-032	2 (2 x 1)	1.510 (38.35)	"	1.530 (38.86)	"	2V2H	Yes
-023	-033	3 (3 x 1)	2.265 (57.53)	"	2.285 (58.04)	"	2V2H	Yes
-024	-034	4 (4 x 1)	3.020 (76.71)	"	3.040 (77.22)	"	2V2H	Yes
-025	-035	5 (5 x 1)	3.775 (95.89)	"	3.795 (96.39)	"	2V2H	Yes
-026	-036	6 (6 x 1)	4.530 (115.06)	"	4.550 (115.57)	"	2V2H	Yes
-027	-037	7 (7 x 1)	5.285 (134.24)	"	5.305 (134.75)	"	2V2H	Yes
-028	-038	8 (8 x 1)	6.040 (153.42)	"	6.060 (153.92)	"	2V2H	Yes
-029	-039	9 (9 x 1)	6.795 (172.59)	"	6.815 (173.10)	"	2V2H	Yes
-030	-040	10 (10 x 1)	7.550 (191.77)	"	7.570 (192.28)	"	2V2H	Yes
-041	-051	2 (1 x 2)	.755 (19.18)	1.510 (38.35)	.775 (19.69)	1.530 (38.86)	2V	No
-042	-052	4 (2 x 2)	1.510 (38.35)	"	1.530 (38.86)	"	2V	No
-043	-053	6 (3 x 2)	2.265 (57.53)	"	2.285 (58.04)	"	2V	No
-044	-054	8 (4 x 2)	3.020 (76.71)	"	3.040 (77.22)	"	2V	No
-045	-055	10 (5 x 2)	3.775 (95.89)	"	3.795 (96.39)	"	2V	No
-046	-056	12 (6 x 2)	4.530 (115.06)	"	4.550 (115.57)	"	2V	No
-047	-057	14 (7 x 2)	5.285 (134.24)	"	5.305 (134.75)	"	2H	No
-048	-058	16 (8 x 2)	6.040 (153.42)	"	6.060 (153.92)	"	2H	No
-049	-059	18 (9 x 2)	6.795 (172.59)	"	6.815 (173.10)	"	2H	No
-050	-060	20 (10 x 2)	7.550 (191.77)	"	7.570 (192.28)	"	2H	No

See footnotes at end of table.

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Ⓐ TABLE III. Dash number, design and panel cutout dimensions - Continued.

Dash number 1/		Number of stations (H x V)	Dimensions 2/				Number of flanges 3/	RFI shielded
			Matrix ±.015 (.38)		Panel cutout +.030 (.76) -.000			
Type S	Type W		A	B	C	D		
-061	-071	2 (1 x 2)	.755 (19.18)	1.510 (38.35)	.775 (19.69)	1.530 (38.86)	2V2H	Yes
-062	-072	4 (2 x 2)	1.510 (38.35)	"	1.530 (38.86)	"	2V2H	Yes
-063	-073	6 (3 x 2)	2.265 (57.53)	"	2.285 (58.04)	"	2V2H	Yes
-064	-074	8 (4 x 2)	3.020 (76.71)	"	3.040 (77.22)	"	2V2H	Yes
-065	-075	10 (5 x 2)	3.775 (95.89)	"	3.795 (96.39)	"	2V2H	Yes
-066	-076	12 (6 x 2)	4.530 (115.06)	"	4.550 (115.57)	"	2V2H	Yes
-067	-077	14 (7 x 2)	5.285 (134.24)	"	5.305 (134.75)	"	2V2H	Yes
-068	-078	16 (8 x 2)	6.040 (153.42)	"	6.060 (153.92)	"	2V2H	Yes
-069	-079	18 (9 x 2)	6.795 (172.59)	"	6.815 (173.10)	"	2V2H	Yes
-070	-080	20 (10 x 2)	7.550 (191.77)	"	7.570 (192.28)	"	2V2H	Yes

1/ Dash numbers shown are for matrix frame with flanges and receptacles and do not include switch, indicator, or lens assemblies. See MIL-S-22885/86.
2/ Metric equivalents are given for general information only.
3/ See figure 3 (2V = two vertical flanges; 2H = two horizontal flanges).

Ⓐ CONCLUDING MATERIAL

Custodians:

Navy - EC
Air Force - 85

Review activities:
Navy - AS, SH
Air Force - 99

Preparing activity.
Navy - EC

Agent:
DLA - ES

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